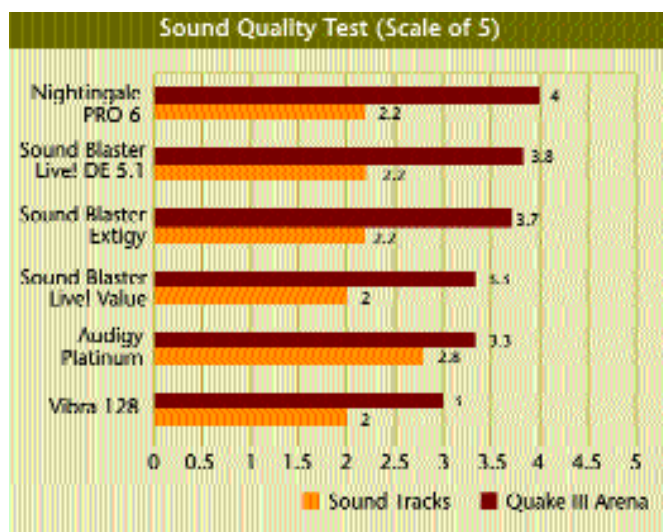
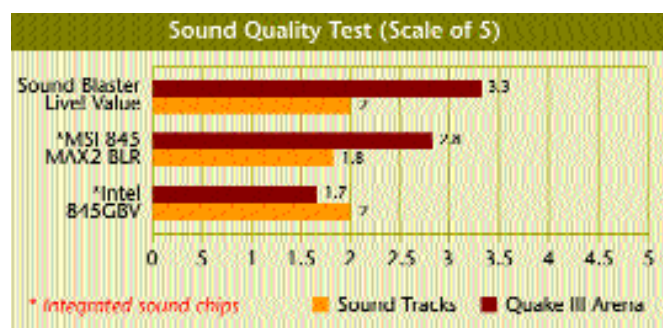




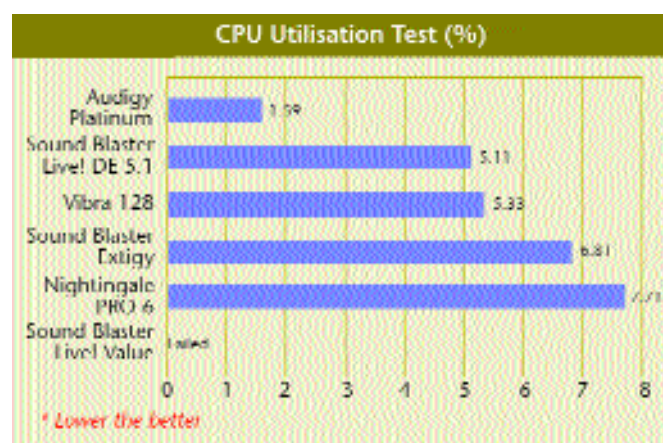
subjective analysis was carried out to determine the quality of sound that a card can deliver under various applications. To ascertain the validity of the technical claims that a manufacturer makes for its product, we used a very comprehensive evaluation tool—RightMark Audio Analyzer (RMAA). We also used a game and a benchmarking suite to evaluate the stress that a card puts on the overall computer system. Thus, by choosing this mix of subjective and analytical tests, we were able to obtain a clear picture of the performance of a soundcard.

Sound quality: To get a handle on the performance of a soundcard while playing music, we used an audio CD that had a mix of Hindi and English songs. This mix was chosen for both the variation in the musical instruments used and the fact that Indian songs tend to be more vocal-heavy than their Western counterparts. Apart from the clarity of each instrument, overall bass and treble output were given importance while evaluating the cards. The tracks were played to three different reviewers who rated each on a scale of 5; the weighted averages of the scores were then logged.

The scores of all the soundcards ranged between 1.9 and 2.8 (out of a maximum of 5). This was a poor showing on the part of all the contenders. The Audigy did win, but with a paltry score of 2.8 out of 5. The other cards barely crossed the mark of 2 points. The Creative Sound Blaster Live! DE 5.1 came in second with a score of 2.2, but it produced a slight crackling sound and its bass output was also not up to the mark.



Integrated sound chips have evolved to a level where they can closely compete with the best soundcards in the market and offer an ideal solution for the cash-strapped consumer



We also accessed the sound quality of a card while playing games. A quick fragfest under *Quake III Arena* (map DM9) left our reviewers with grins on their faces. This game has amazing sound effects and a very funky music score that consists of some cool guitar riffs and drum beats. We observed the clarity with which a card presented these effects. Moreover, the game engine's use of Doppler sound effects (for the rocket launcher) also helped us evaluate the effectiveness of a cards' multi-channel support. Amongst the cards, the Zoltrix Nightingale PRO 6 impressed us the most with a score of 4 out of the maximum of 5. With 5.1 speaker support, the Nightingale brought every fight to life, even as it offered a thumping bass and an amazingly clear treble. The Creative Sound Blaster Live! DE 5.1 came in a close second with 3.8 points, whereas the Creative Vibra 128 lagged behind the rest of the pack with a below-average score of 3 points.

CPU utilisation: A soundcard takes up precious processor resources while doing its job. This has repercussions in the multi-tasking environment that current operating systems deliver, as task-switching can become a painful experience. Needless to say, a low number in this test indicates a good performance by the soundcard. We used 3D Audio Winbench, an application that gives the total CPU usage by a soundcard. The results were quite interesting, with the Creative Audigy Platinum topping this test with an excellent score of 1.59 per cent utilisation. Surprisingly, the Sound Blaster Live! Value failed this test, despite updating its drivers. The external Sound Blaster Extigy was plagued by the limitations of its USB connection and posted the second worst score of 6.81 per cent. The worst score however, belonged to the Zoltrix Nightingale PRO 6 (7.71 per cent). A thing to keep in mind while digesting all this is that current processors are speed demons. If the card consumes 7 per cent of resources out of a total available speed of say, 1 GHz, you won't notice any difference in speed.

This became all the more apparent in our game tests using *Quake III Arena*. We recorded the fps under both Normal and High Quality sound mode. By doing this, we hoped to bring to light the overhead that high-quality sound puts on a system. We were expecting notable differences in the score under the two modes—the fps should have been lesser while running the game with high quality sound enabled. The results belied our expectations—the scores under both the modes and across all the cards (except for the Creative Sound Blaster Extigy) did not vary by much. The performance of the Extigy solution can again be blamed on its USB 1.1 connection. So, if you have a high-end processor in your system, you can be rest assured